

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458658.771631 +/- 3.3e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.016
Transit depth (Rp/Rs)^2	2.66 +/- 0.52 [%]
Orbital Inclination (inc)	89.96 +/- 0.39
Airmass coefficient 1 (a1)	1.274 +/- 0.018
Airmass coefficient 2 (a2)	-0.223 +/- 0.066
Scatter in the residuals of the lightcurve fit is	1.4 %
Best Comparison Star	None
Optimal Aperture	4.32
Optimal Annulus	5.4
Transit Duration (day)	0.1213 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.016 vs 0.1356 (deviation 1.18σ)
Duration fit vs published	0.1213 d vs 0.1129 d (deviation 3.05σ)
Mid-time O–C	-0.4 min
Depth SNR	7.0
Residual scatter	1.4 %

Light curve

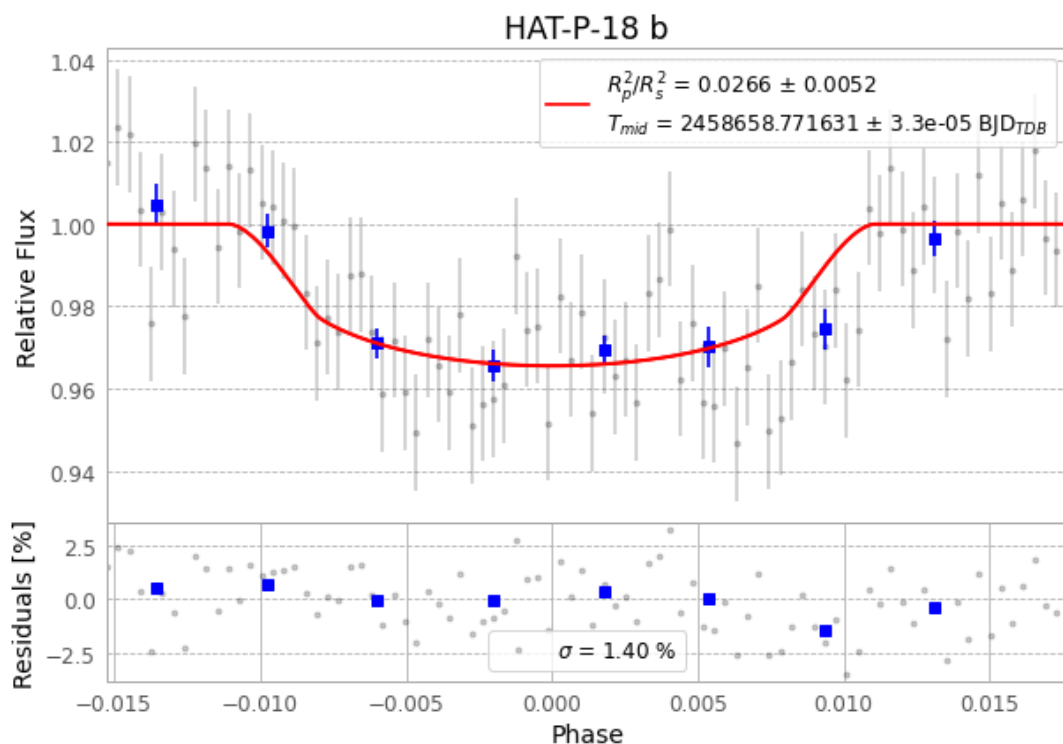


Figure 1 — FinalLightCurve_HAT-P-18 b_2019-06-23.png (SHA-256 ec74a28e678491c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bb41bdc749cb7979...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

88 FITS frames (58 MB), HATP-18190624042412.FITS → HATP-18190624084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-190624.log (f33bc0385700...), FinalLightCurve_HAT-P-18 b_2019-06-23.png (ec74a28e6784...), FinalLightCurve_HAT-P-18 b_2019-06-23.csv (3d3bdf3c0a95...)

Compute resources

Wall time 138.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:06Z.